

23CSE502

Software Testing

Module-1 Testing Principles and Axioms

Testing as a Process - Testing Axioms - Software Testing Principles - Origins and Cost of Defects - Defect Classes and Examples - Developer/Tester Support of Developing a Defect Repository - Defect Prevention Strategies.

Testing Axioms

- The test axioms are a context neutral set of rules for testing that identify the critical thinking process and motivations for all test approaches.
- The axioms represent mechanism designed to meet the objective of the testing in scope.
- A mechanism may be well defined, documented process, an informal or even ad-hoc activity - but that mechanism must be understood and used by participants in the test.

Axiom

An axiom – A sentence or proposition that is not proved and considered as obvious for theory building and acceptance

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Testing Axioms

1. **THE STAKEHOLDER AXIOM:** testing needs stakeholders
2. **THE VALUE AXIOM:** the value of intelligence is independent of who produce it
3. **THE SCOPE MANAGEMENT AXIOM:** if we don't manage scope, we may never meet stakeholder expectations
4. **THE GOOD-ENOUGH AXIOM:** the scope of testing and acceptance are always compromises
5. **TEST BASIS AXIOM:** testers need sources of knowledge to select things to test

Testing Axioms

- 6. THE ORACLE AXIOM:** testers need sources of knowledge to evaluate actual outcomes or behaviours
- 7. THE COVERAGE AXIOM:** testing needs a test coverage model or models
- 8. THE PRIORITISATION AXIOM:** prioritizing the tests needs a mechanism for ordering tests by value
- 9. THE FALLIBILITY AXIOM:** our sources of knowledge are fallible and incomplete (Ready for retesting)
- 10. THE REPEAT-TEST AXIOM:** some repeated tests are unavoidable

Testing Axioms

- 11. THE EXECUTION SEQUENCING AXIOM:** run our most valuable tests first – we may not have time to run them later
- 12. THE ENVIRONMENT AXIOM:** test execution requires a known, controlled environment
- 13. THE EVENT AXIOM:** testing never goes as planned; evidence arrives in discrete quanta
- 14. The design axiom:** Test design is based on models
- 15. The delivery axiom:** Intelligence produced by test determines the value of testing

Testing Axioms:

Note:

1. It is impossible to test a program completely.
2. Software testing is a risk-based exercise.
3. Testing cannot show the absence of bugs.
4. The more bugs you find, the more bugs there are.
5. Not all bugs found will be fixed.
6. It is difficult to say when a bug is indeed a bug.
7. Specifications are never final.
8. Software testers are not the most popular members of a project.
9. Software testing is a disciplined and technical profession.